

generality makes it challenging to design a general model of knowledge-level connections among ML models. On the other hand, model linking should be non-intrusive to the original inference system and require as little model information and code modification as possible. The black-box access of ML models bringing additional challenges to the design and implementation.

(2) **How to efficiently select models to be executed and models to be predicted?** Given a set of ML models, after constructing model links among them, we need to select a proper subset of models to be executed under a certain cost budget, e.g. the allocatable GPU memory. Highly efficient model selection is critical to the cost-performance trade-off, which is non-trivial due to its theoretical hardness.

In this paper, we first formalize the model linking task and propose the design of model links which supports linking heterogeneous black-box ML models. And we develop a model link-based algorithm, named *MLink*, to schedule multi-model inference under a cost budget. We evaluated our designs on a multi-modality dataset with seven different ML models, covering five classes of learning tasks and three types of input modalities. Results show that our proposed model links can be effectively built among heterogeneous black-box models. We evaluated *MLink* on two real-world video analytics systems, one for the smart building and the other for city traffic monitoring, including six visual models and 3,264 hours of video from 58 cameras. Under the budget of GPU memory, *MLink* outperforms baselines (multi-task learning (Crawshaw 2020), deep reinforcement learning-based scheduler (Yuan et al. 2020a) and frame filtering (Li et al. 2020)) and can save 66.7% inference computation while preserving 94% output accuracy.

Problem Statement

In this section, we define the model linking task and the inference under budgets problem.

Model linking. Given a set of black-box ML models $F = \{f_i\}_{i=1}^k$, where $f_i : X_i \rightarrow Y_i$ is a function mapping the input to its inference result. ML models can be highly heterogeneous, i.e., different input modalities, learning tasks, architectures, etc. We only assume that input spaces $\{X_i\}_{i=1}^k$ are the same or aligned. The case that different models share the same input spaces is common, e.g., multi-task learning-based robotics (Crawshaw 2020; Zhang and Yang 2021) and multimedia advertising (Yuan et al. 2020b). The aligned input spaces typically exists in the context of multi-modal scenarios, e.g., multi-modality event detection (Elhoseiny et al. 2016) and visual speech synthesis (Baltrušaitis, Ahuja, and Morency 2018). In practice, synchronization in time can easily align inputs for many applications. Moreover, approaches such as spatial alignment of multi-view videos (Black, Ellis, and Rosin 2002) and audio-visual semantic alignment (Wang, Fang, and Zhao 2020) can be adopted for specific scenarios. We define model linking as a function $g_{ij} : Y_i \rightarrow Y_j$, i.e., a mapping from the source model f_i 's output space to the target model f_j 's. Then the composite function $g_{ij} \circ f_i : X_i \rightarrow Y_j$ can perform the inference computation of f_j . Correspondingly, g_{ji} links the knowledge of f_i into $g_{ji} \circ f_j$.

Multi-source model links ensemble. When the number of models $k \geq 3$, for one target model f_j , there could be multiple model links from different sources. Let $A \subseteq F$ denote the set of source models. Then for all $f_i \in A$, $g_{ij} \circ f_i$ performs the prediction task to f_j 's inference outputs. The question that follows is, how do we determine the final prediction? From the ensemble learning perspective, $\{g_{ij} \circ f_i\}_{f_i \in A}$ constitute a multi-expert model (Yuksel, Wilson, and Gader 2012), which has the potential to perform better prediction with the multi-task & multi-modal representation (Baltrušaitis, Ahuja, and Morency 2018; Zhang and Yang 2021). We define $h_{A,j}$ as the ensemble model link from A to f_j . So the input of $h_{A,j}$ is the set of predictions by g_{ij} where $f_i \in A$. Note that if A has only one element f_i , then $h_{A,j} = g_{ij} \circ f_i$.

Multi-model inference under budget. The model links can be utilized to achieve resource-performance trade-offs of multi-model inference workloads. Let $c(\cdot)$ denote the cost of running a function, e.g., GPU memory or inference time. For resource-limited devices (e.g., smartwatches and mobile phones) and delay-sensitive tasks (e.g., real-time video analytics and audio assistant), there are certain constraints on the total cost. We define B as the cost budget and aim to maximize the inference accuracy under that budget. Let $p(h_{A,j})$ denote the performance measure of the ensemble model link, which depends on the target model's task. We assume the range of p is normalized into $[0, 1]$. For example, the performance measure can be accuracy for classification task and bounding box IoU for the detection task. Following previous efforts for optimizing the inference efficiency (Li et al. 2020; Yuan et al. 2020a), the performance measure the consistency between obtained results and exact inference outputs, instead of ground-truth labels. The multi-model inference under cost budget problem is formalized as:

$$\begin{aligned} & \max_{A \subseteq F} \left(\overbrace{\frac{1}{|F|} \left(\underbrace{\sum_{f_i \in A} 1}_{\text{activated}} + \underbrace{\sum_{f_j \in F \setminus A} p(h_{A,j})}_{\text{predicted}} \right)}^{\text{average output accuracy}} \right) \\ & s.t. \quad \underbrace{\sum_{f_i \in A} c(f_i)}_{\text{exact inference}} + \underbrace{\sum_{f_j \in F \setminus A} c(h_{A,j})}_{\text{model links}} \leq B. \end{aligned} \quad (1)$$

Under the cost budget, the optimization problem aims to maximize the average performance of all models F by selecting an *activated* subset A to be executed. For ease of description, we define the objective function as the *output accuracy*. Activated models do exact inference, so their performance scores are all 1. Models that are not activated only participate in constructing model links and will not be executed during the inference phase; instead, they are predicted by the activated models via ensemble model links. The cost of activated models is performing exact inference, while the cost of predicted models is from running model links. So the model links should be both accurate and lightweight to reduce the cost while preserving the quality of the multi-model inference workloads.